Federal Reserve Sentiment Model

Google Colab Code:
https://colab.research.google.com/drive/1MGdXKKZAq5B7IrSJfiZ9J9L5X1sr-NGe#scrollTo=My4krUcaxAJL

GitHub: https://github.com/cpsc1710-fa25/1710-final-project-federal-reserve-sentiment-model

Introduction

This project develops a quantitative model to measure the hawkish and dovish tone of Federal Reserve Chair Jerome Powell's FOMC press conferences from 2018 to 2025. Central bank communication is one of the most influential forces shaping financial markets, specifically the Chair's speech after Federal Open Market Committee (FOMC) meetings after policy decisions. The 30-45 minute press conference after monetary policy decisions get analyzed and then analyzed again by economists, market participants and people all over the world more generally. Oftentimes wrong conclusions can be made on the intent of what and how the Chair is trying to frame communication for future monetary policy decisions. By using LLM and transformer based infrastructure, we can quantitatively gauge the sentiments of the Chairmen's intent!

Generic sentiment models often misclassify monetary policy statements because they cannot distinguish technical economic language from typical positive or negative sentiment. To address this gap, the objective of this project is to build a hybrid model that integrates a domain-specific lexicon of monetary policy signals with a pretrained financial sentiment model, FinBERT. The resulting system generates meeting-level policy stance scores that track how the tone of Fed communication has evolved through major macroeconomic regimes. The aim is to create a tool that extracts meaningful structure from central bank communication and provides a more objective, data-driven measure of policy stance.

Methods and Results

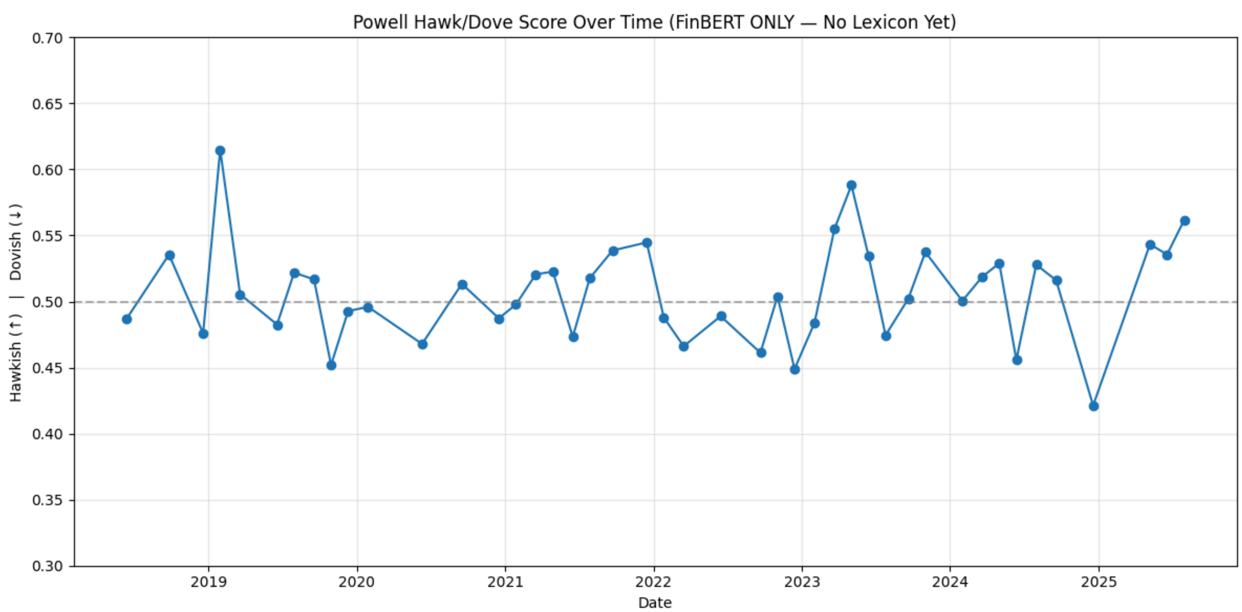
The transcripts of Powell's FOMC press conferences were collected, cleaned, and split into individual sentences to allow for fine-grained analysis. Each sentence was processed through FinBERT, a financial sentiment classifier that outputs probabilities of positive, neutral, and negative tone. We used an aggregate of 44 speeches from Chair Powell throughout 2018 - 2025, after FOMC meetings on monetary policy.

FinBERT is a transformer-based language model built on the BERT architecture and fine-tuned specifically for financial text. Unlike generative large language models such as GPT, FinBERT is an encoder-only model designed to classify sentiment rather than produce new text. It learns contextual representations of sentences by reading them bidirectionally, which allows it to detect subtle positive, negative, or neutral tones in financial language. GPT models are decoder-only and used for next token prediction.

FinBERT performs well in domains involving earnings releases, market commentary, and financial news because it was trained on datasets for these contexts. However, it was not built to interpret central bank communication, where terms like “slowing growth,” “tight labor market,” or “muted inflation” carry policy meanings that do not align with typical market sentiment labels.

In practice, the two main areas it incorrectly predicts are: growth and labor markets. In central bank language, slower growth is counterintuitively good because it means a more “dovish” or accommodative central bank (and vice versa). Similarly, weaker labor markets and increasing unemployment is bad for the economy but “dovish” for central banks. Dovish sentiment by the central bank can paradoxically actually be good for markets! This is why sometimes you will see slower growth and weaker labor markets actually drive market prices higher. The mechanism is due to to a more dovish Federal Reserve and lower interest rates.

Below is an output using the FinBERT model alone. We can see there is no real trend in this data. Additionally, one of the most dovish speeches, January 2019, in our final model is the most “Hawkish” (tighter monetary policy) in the model below! There is also no clear trend and oscillation throughout the period which shouldn’t be the case given the economic dynamics and monetary policy during the period.



Again, we know off the bat this is incorrect, but what is exactly going on? The initial misclassification occurs because FinBERT, although trained on financial text (like we mentioned above), does not understand the Federal Reserve’s domain-specific vocabulary or the way central bankers discuss growth, employment, and inflation. In its training data, negative language about the economy is again typically interpreted as bad news for markets and therefore labeled as negative sentiment. But in monetary policy communication, these same phrases have almost the opposite meaning! To fix this, we added an extensive list of phrases and words to look out for to override FinBERT.

Below are some of the manual phrases we added to the FinBERT model to improve the models accuracy. There is a wide variety to help catch Federal Reserve specific speech.

Add corrections to FinBERT model

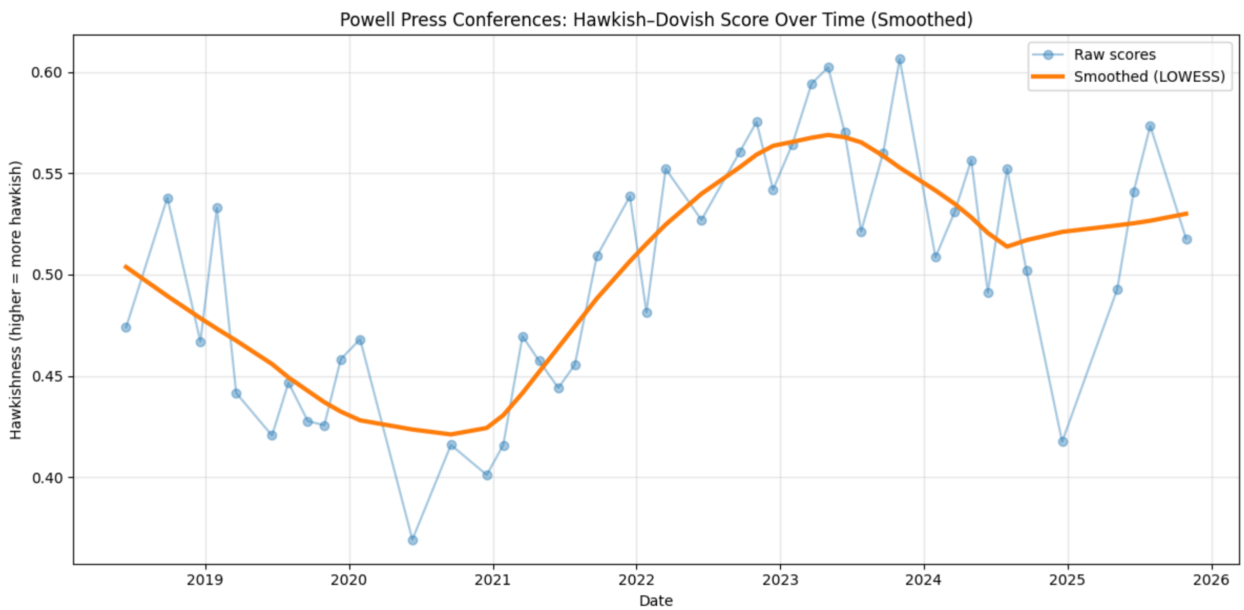
```
# ---------- (A) Weighted hawkish/dovish lexicon ----------
HAWKISH_PATTERNS = {
    # --- Restrictive policy / stance language ---
    r"\brestrictive (stance|policy)\b": 2.0,
    r"\bremain(s|ing)? restrictive\b": 1.8,
    r"\bpolicy (is|remains) (appropriately )?restrictive\b": 1.9,
    r"\bmaintain a restrictive stance\b": 2.0,
    r"\brestrictive levels of policy\b": 1.8,

    # --- Inflation too high / price pressures ---
    r"\binflation (is|remains|continues to be) (too|unacceptably) high\b": 2.2,
    r"\belevated inflation\b": 1.6,
    r"\binflationary pressures\b": 1.7,
    r"\bprice pressures?\b": 1.4,
    r"\bpersistent inflation\b": 1.8,
    r"\binflation remains elevated\b": 1.9,
    r"\bupside inflation risks\b": 1.8,
    r"\binflation has not come down\b": 1.8,
    r"\bprogress on inflation has been slow\b": 1.7,

    # --- Commitment to price stability ---
    r"\brestore(ing)? price stability\b": 2.2,
    r"\bbring inflation down\b": 1.5,
    r"\bwe are strongly committed\b": 1.6,
    r"\bwe will keep at it\b": 1.7,
    r"\b(unconditional|overarching) (focus|commitment)\b": 1.6,

    # --- Further tightening guidance ---
    r"\bfurther tightening\b": 2.0,
    r"\bfurther rate increases?\b": 1.9,
    r"\badditional tightening\b": 1.8,
    r"\btightening may be appropriate\b": 1.8,
    r"\bpolicy may need to tighten (further)?\b": 1.7,
    r"\bmore tightening ahead\b": 1.7,
    r"\braise (rates|the policy rate)\b": 1.9,
    r"\bcontinue to raise\b": 1.8,
    r"\bwe expect to raise\b": 1.8,

    # --- No cuts / not done yet ---
    r"\bnot (yet )?sufficiently restrictive\b": 1.9,
    r"\bwe have more work to do\b": 1.7,
    r"\bnot confident inflation is moving down\b": 1.7,
    r"\bit is too early to consider cutting\b": 1.7,
    r"\bnot considering rate cuts\b": 1.7,

    # --- Labor market / overheating ---
    r"\boverheating labor market\b": 1.8,
    r"\btight labor market\b": 1.5,
    r"\bvery strong labor market\b": 1.4,
    r"\bexcess demand\b": 1.6,
    r"\blabor market remains tight\b": 1.5,
```

```
DOVISH_PATTERNS = {
    # --- Your existing patterns ---
    r"\baccommodative (stance|policy)\b": 1.6,
    r"\bsupport (the )?recovery\b": 1.2,
    r"\beasing (financial )?conditions\b": 1.2,
    r"\bprogress on disinflation\b": 1.3,
    r"\bcloser to (our )?target\b": 1.2,
    r"\bdownside risks to growth\b": 1.1,

    # --- Strongly dovish “Powell Pivot” language (Jan 2019) ---
    r"\bpatient\b": 1.0,
    r"\bwe (will )?be patient\b": 1.0,
    r"\bwait and see\b": 1.4,
    r"\bno preset (path|course)\b": 1.5,
    r"\bnot on a preset (path|course)\b": 1.5,
    r"\bmeeting by meeting\b": 1.3,
    r"\bdata(-| )dependent\b": 1.2,

    # --- Weak inflation = dovish ---
    r"\bmuted inflation\b": 1.7,
    r"\binflation pressures (are )?(muted|subdued|benign)\b": 1.1,
    r"\b(inflation|price pressures) (running )?below (our )?target\b": 1.

    # --- Global risk-off / downside risk ≠ hawkish ---
    r"\bdownside risks\b": 1.3,
    r"\bglobal (economic )?(slowdown|weakness|headwinds)\b": 1.3,
    r"\bslowing global growth\b": 1.2,
    r"\bslowing growth\b": 1.2,

    # --- When tightening conditions LOWER the odds of hikes ---
    r"\bfinancial conditions (have )?tightened\b": 1.2,
    r"\btightening financial conditions\b": 1.2,

    # --- Expansion support language (classic dovish) ---
    r"\bsustain the expansion\b": 1.4,
    r"\bsustain this expansion\b": 1.4,
    r"\bsupport the expansion\b": 1.3,
    r"\bas appropriate to sustain\b": 1.3,
    r"\bsupport the economic recovery\b": 1.3,

    # --- “Ready to ease if needed” ---
    r"\bstand ready to use our tools\b": 1.3,
    r"\bif the economy (were|was) to weaken\b": 1.3,
    r"\bif downside risks (were|are) to materialize\b": 1.3,

    # --- Low neutral rate / limited hiking room ---
    r"\blower neutral (rate)?\b": 1.2,
    r"\bclose to estimates of the longer-run level\b": 1.3,

    r"\bbe patient\b": 1.7,
    r"\bpatient approach\b": 1.5,
    r"\bpatient wait\b": 1.5,
```

Once accounting for the Lexicon update, we have a new and smoother model which actually maps itself neatly to past data. We can see it follows what happened in real life quite nicely. Below is a recounts the events

- **2018:** The model shows a strong hawkish tone reflecting the fiscal stimulus and tax cuts under the Trump administration, alongside the Fed raising rates for the first sustained period since before the 2008 crisis. It is not as hawkish as we might expect, but it still captures the general tightening stance.
- **2019:** Powell's January 2019 pivot was distinctly dovish, but our earlier model incorrectly labeled it as hawkish because many non-informative sentences were misclassified by FinBERT. This was one of the core issues that motivated building a lexicon- and stance-based correction.
- **2020–2021:** The model correctly captures peak dovishness during the pandemic as the Fed cut rates to zero, expanded balance sheet policies, and implemented broad liquidity support.

- **2022–2023:** With inflation rising to multi-decade highs, the model shows a sharp hawkish shift, reflecting 75-basis-point hikes and the aggressive tightening cycle.
- **2024–2025:** As inflation cools and the labor market softens, the model shows a recalibration toward a more balanced or slightly dovish stance, consistent with the Fed preparing for potential easing.



<u>January 2019 Powell Pivot:</u>
What we can see below is the clear point where Powell turns from Hawkish to Dovish! He explicitly says “after developments over the last several months warrant a patient, wait-and-see approach regarding future policy changes”. However, even adjusting for our own lexicon the model still views January 2019 as hawkish (albeit to a smaller extent than FinBERT alone). I focus on this data point because other than the COVID period, this is one of the most “Dovish” speeches in Jay Powell’s term.

—

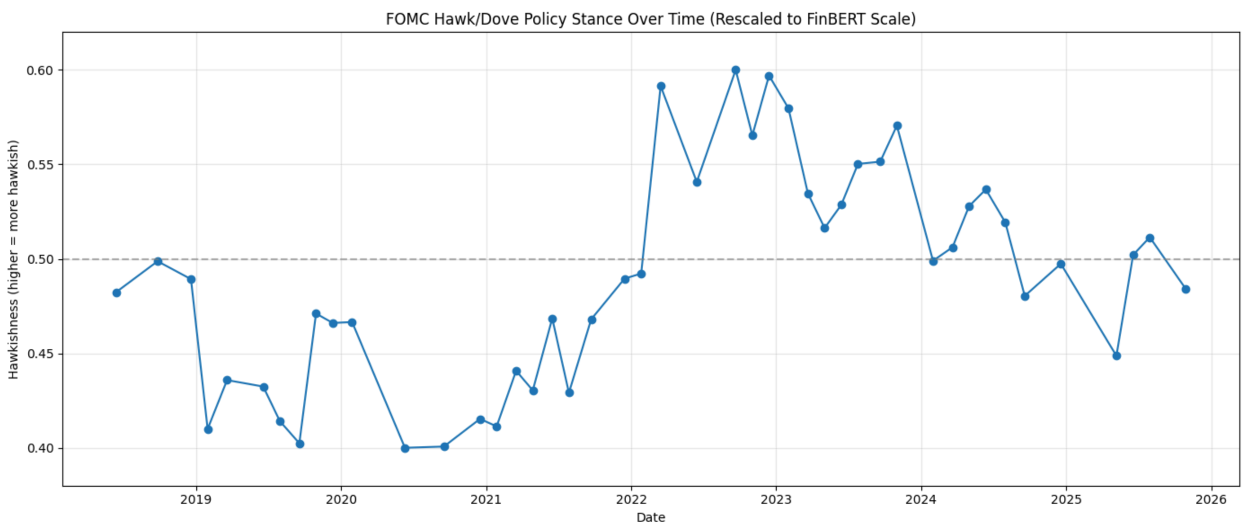
> At our December meeting, we noted the solid outlook for steady growth, vigorous job creation, and price stability. We also stressed that the extent and timing of any rate increases were uncertain and would depend on incoming data and the evolving outlook. We therefore said that we would be paying close attention to global economic and financial developments and assessing their implications for the economic outlook.
>
> Today the FOMC decided that the cumulative effects of those developments over the last several months warrant a patient, wait-and-see approach regarding future policy changes. In particular, our statement today says, "In light of global economic and financial developments and muted inflation pressures, the Committee will be patient as it determines what future adjustments to the target range for the federal funds rate may be appropriate."

This model actually does a decent job at predicting what is going on at the Fed, but it was still getting the directionality of specific meetings incorrect (like January 2019). While this was an interesting model, ex-ante we do not know if it will treat random sentences as hawkish with no explanatory value! Below are the top Hawkish sentences from the FinBERT model. 90% of these have any explanatory value! They mention page numbers, members of media, or are blanket statements.

doc_id	sent_	text	hawk_f
FOMCpresconf20200610	65	We tabulate those submissions, and we publish them as the SEP.	0.934781
FOMCpresconf20201216	38	As we previously announced, we are now releasing the entire package of our SEP r	0.928454
FOMCpresconf20200916	78	Does the guidance today apply to the current asset purchase pace?	0.92668
FOMCpresconf20190619	85	Chairman, Steve Liesman, CNBC.	0.926035
FOMCpresconf20181219	109	Chairman, Steve Liesman, CNBC.	0.926035
FOMCpresconf20230614	540	A, a large part of it is in smaller banks.	0.925876
FOMCpresconf20230614	189	It's job Page 10 of 26 June 14, 2023 Chair Powell's Press Conference FINAL creation	0.925163
FOMCpresconf20210616	156	The timing, of course, Ylan, will depend on the pace of that progress and not on a	0.924718
fomcpresconf20250507	363	We will in Page 15 of 25 May 7, 2025 Chair Powell's Press Conference FINAL six wee	0.924674
FOMCpresconf20191030	168	Page 9 of 24 October 30, 2019 Chair Powell's Press Conference FINAL NICK TIMIRA	0.924505
FOMCpresconf20190130	385	We Page 19 of 24 January 30, 2019 Chairman Powell's Press Conference FINAL only	0.924501
FOMCpresconf20210922	170	So all of these everyone's, you know, ownership and activities are all disclosed or	0.924488
FOMCpresconf20190130	458	Chairman Powell, Myles Udland with Yahoo Finance.	0.923753
FOMCpresconf20180613	99	Page 6 of 26 June 13, 2018 Chairman Powells Press Conference FINAL NICK TIMIRA	0.923753
FOMCpresconf20210428	505	Page 24 of 29 April 28, 2021 Chair Powells Press Conference FINAL The other one is	0.922893
FOMCpresconf20211215	548	We'll have another at the next meeting and another at the Page 26 of 31 December	0.922684
FOMCpresconf20210317	337	And, you know, we conduct policy, Page 16 of 30 March 17, 2021 Chair Powellâ€™s	0.922652
FOMCpresconf20180926	283	Chairman, Michael McKee, Bloomberg Radio and Television.	0.922552
FOMCpresconf20200610	513	Chairman, Michael McKee, Bloomberg Television and Radio.	0.922421

To update this, we had to do another iteration and tell the model to only give values when our custom Lexicon was hit. This dramatically improved directionality for the model.

There still are some incorrect statements being judged with wrong scores, but a majority of them are actually directionally the correct way! When this is done the model represents actual Federal Reserve Sentiment nicely. We scale below to from 0.4 to 0.6 to match the previous charts.



The model uses 1/3 of each of our signals: Lexicon, Stance, FinBERT. Stance + Lexicon is for directionality. Lexicon is for intensity, FinBERT is the broader sentiment.

Model Evaluation

A good test against our model is using other sources models to dictate how wells ours works. We use 3 models below: 1) Bloomberg, 2) the actual Federal Reserve, and 3) ChatGPT 5.1 Thinking Mode. Below are the outputs of each of the models.

Because we can't get time series data from all of the sources we have to go more qualitatively unfortunately versus quantitatively.

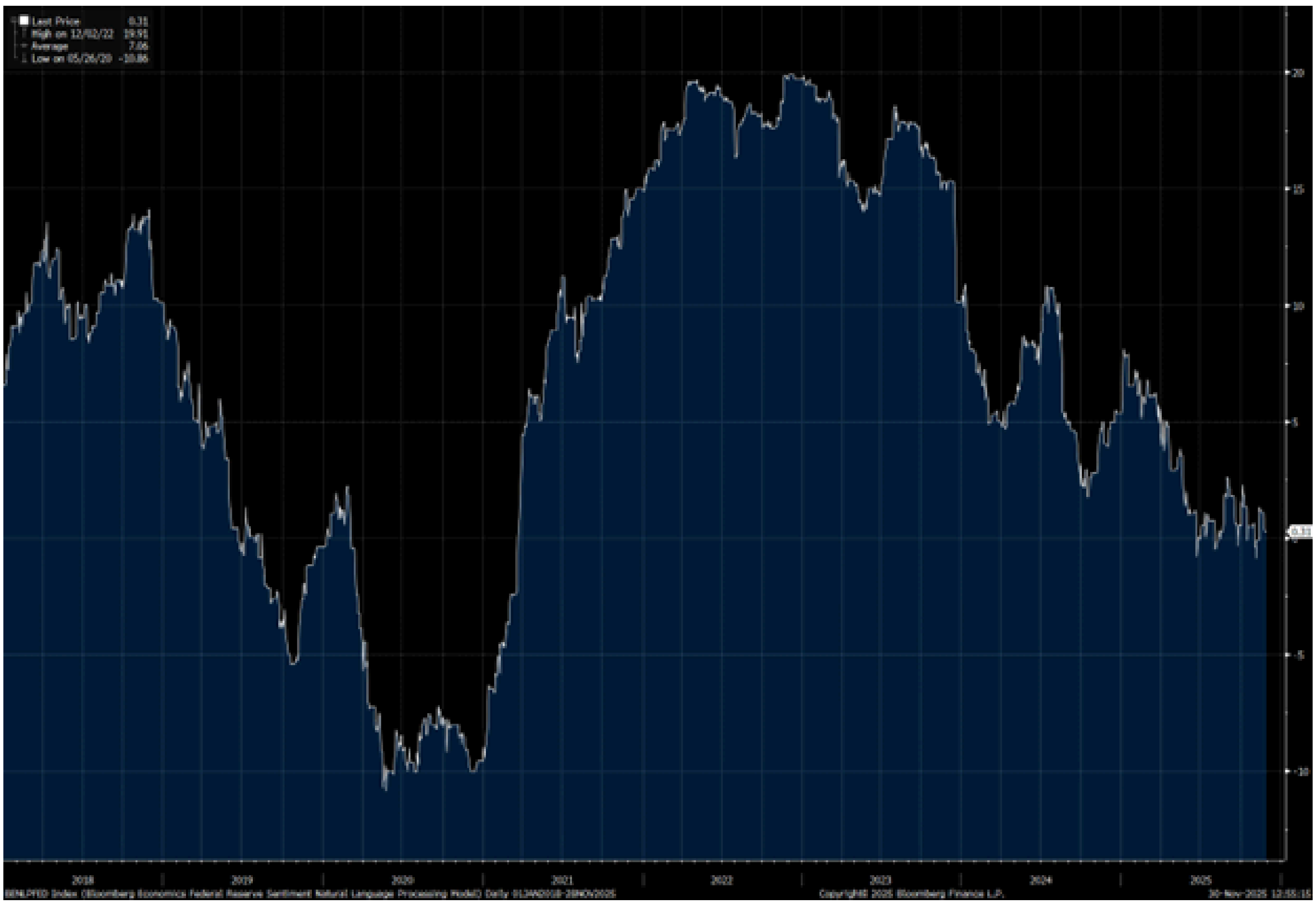
Compared with our model, the Bloomberg index moves much more slowly and with broader cycles. While your model shows clear meeting-to-meeting variation (capturing the dovish pivot in early 2019, the plunge during the pandemic, and the steep hawkish turn in 2022) the Bloomberg line tends to evolve over multi-quarter horizons rather than discrete FOMC events. For example, where your model shows a sharp dovish dip in March 2020, the Bloomberg index declines more gradually, reflecting media-driven sentiment adjusting over time. Similarly, Bloomberg's hawkish peak in 2022 occurs with a long buildup, whereas your model shows distinct jumps consistent with explicit Powell communication. The contrast suggests our model is more sensitive to actual Fed speech, while Bloomberg's captures broader market tone rather than specific policy language. This makes sense. All in all the models line up well, with I think Bloomberg doing a better job reflecting 2018 and COVID sentiment than our model.

The academic FOMC sentiment index is more volatile, displaying frequent fluctuations and wider amplitude compared with other models. The chart's swings appear sharper and more abrupt, and it often reacts strongly during recessionary periods highlighted in the graph. This pattern is consistent with a methodology that relies heavily on high-frequency financial news articles and uses a dictionary that treats a wide range of economic terms as sentiment-bearing. Because the underlying data come from thousands of news stories and include many voices beyond the Federal Reserve itself, the resulting index captures both policy communication and the financial press's interpretation of it. The visual result is a signal that is sensitive to fast-moving market narratives but may be less directly tied to the tone of Federal Reserve officials. It is very strange their 2018 sentiment is much more hawkish than 2022; I think our model does a better job calibrating those periods. The overall trend matches well.

The GPT model produces a chart characterized by sharp spikes and large discrete jumps. These swings correspond to the model's reliance on raw counts of hawkish and dovish keywords without contextual understanding. Because the method treats each keyword equally and does not consider the surrounding rhetoric, it tends to amplify meetings in which certain terms happen to appear more frequently (even when the overall tone may not dramatically differ). This results in a line that is reactive and sometimes erratic, with exaggerated highs and lows. The visual pattern highlights a fundamental challenge of dictionary-only approaches: while they are transparent and easy to interpret, they often struggle with contextual interpretation. The GPT model trends similarly but has key spot points which aren't too accurate.

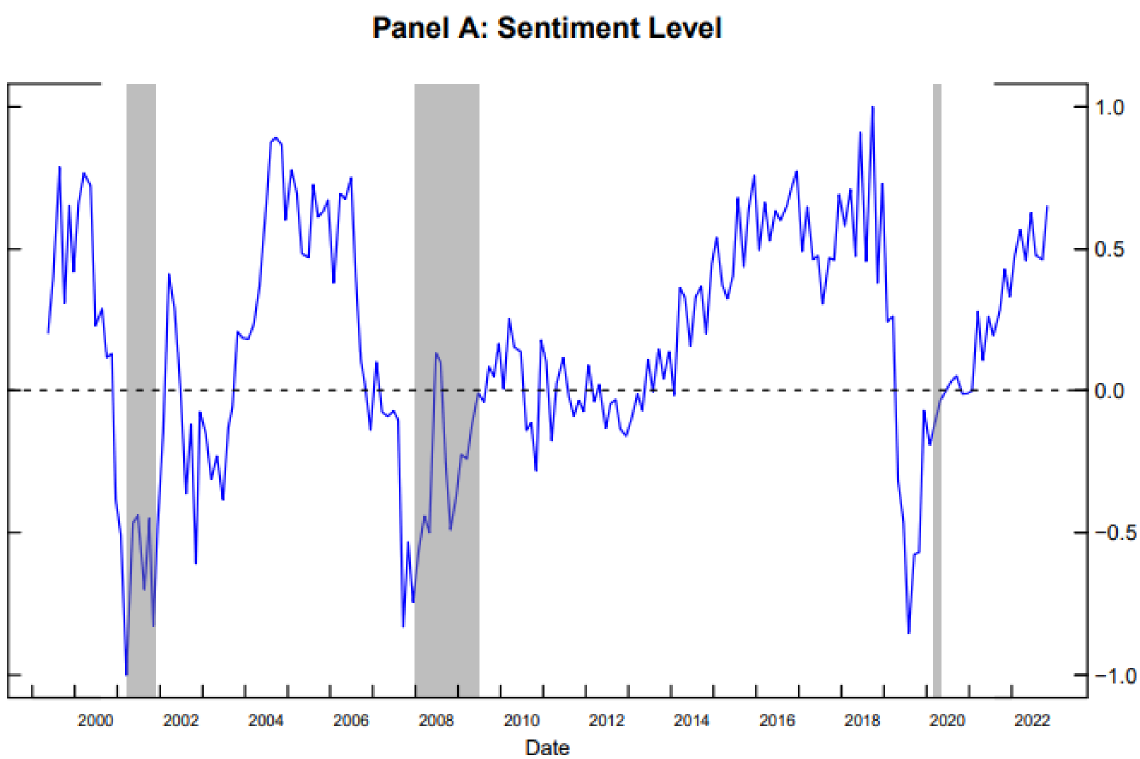
Overall, I would say the model outperforms the FOMC and GPT model, but not the BBG model. The FOMC and BBG model are trained on headlines versus speeches, so they are measuring 2 correlated items but not the same exact thing. The GPT model seems a little too extreme and I think should be taken too seriously.

Bloomberg: The index is underpinned by an NLP algorithm trained on Bloomberg News headlines, covering about 6,200 speaking engagements by Fed officials since 2009. A reading below zero implies rate cuts, while above zero indicates a tightening basis.

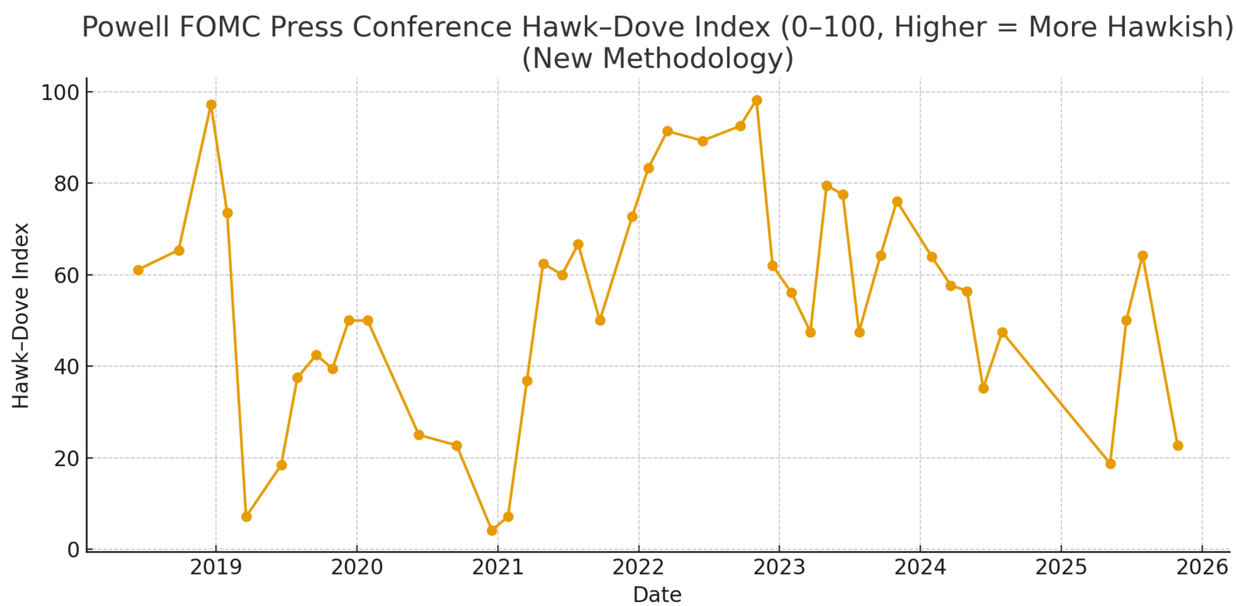


Federal Reserve Open Market Committee Sentiment Model: Apply natural language processing tools to news articles in the financial press to construct a sentiment index—an index of the perceived semantic orientation of monetary policy communications around scheduled Federal Open Market Committee (FOMC) meetings. To that end, we develop several dictionaries that capture various monetary policy tools: conventional monetary policy, asset purchases, and forward guidance. From 1999 - Nov 2022

Figure 5: Sentiment Index and Surprises



ChatGPT 5.1 Thinking (its interpretation on the model): I read each Powell press conference transcript and extracted the full text for that meeting's date. Then I applied a Fed-specific dictionary of clearly hawkish (e.g., "further rate increases," "restrictive stance," "inflation is too high") and dovish (e.g., "disinflation," "downside risks to growth," "accommodative stance") phrases, counting their occurrences to get a net hawk-minus-dove signal for each transcript. Finally, I transformed this net signal into a 0–100 index (0 = extremely dovish, 100 = extremely hawkish) so you can compare stance across meetings over time.



Discussion: Reflections on challenges, lessons learned, broader implications, or potential future work

Developing a model to measure Federal Reserve communication proved more challenging than applying solely standard NLP techniques. A difficulty is that traditional sentiment models, such as FinBERT, are not designed for monetary policy language. Early versions of the model revealed significant misclassification, particularly in 2019 and the pandemic years, because FinBERT interpreted economic phrases through a market-news lens rather than a policy lens. For example, sentences referring to slowing growth or weak labor markets were labeled as negative and therefore hawkish, even though these conditions generally lead the Federal Reserve toward easing. Large portions of each transcript that contained no policy content, such as page markers, speaker names, and Q and A transitions, also contributed to meeting-level sentiment. These problems highlighted the need for domain-specific adjustments.

The first major lesson came from recognizing that central bank communication requires its own vocabulary. Building a weighted lexicon of hawkish and dovish policy terms allowed the model to capture the true intent of statements that FinBERT routinely misclassified. Terms such as further tightening, muted inflation, downside risks, patient, or restrictive stance do not behave like typical financial sentiment, and the lexicon directly injects this expert knowledge into the pipeline. The second lesson involved stance scoring. By introducing a binary stance filter that assigns plus one, minus one, or zero depending on whether a sentence contains substantive policy language, we eliminated a large amount of noise. This step prevented FinBERT from interpreting generic or irrelevant sentences as meaningful policy signals. Together, the lexicon and stance corrections transformed the model from a noisy sentiment classifier into a stable and interpretable policy-stance estimator that aligns well with historical tightening and easing cycles.

The broader implication is that hybrid models combining domain-specific rules with transformer-based NLP are often more accurate and more interpretable than purely neural approaches in specialized fields such as monetary policy. This project demonstrates that financial language models must be adapted carefully when used outside their original contexts. The resulting model is now strong enough to track regime shifts, detect policy pivots, and describe the evolution of Powell's communication style across macroeconomic cycles. With that in mind, this creates an opportunity to fine-tune the model using our own dictionary!

Looking forward, there are two major directions for future work. First, we plan to continue applying and refining this model in real time. The December 10 FOMC meeting will serve as the next test case, and we aim to run the model simultaneously with Powell's press conference. It would take the voice of Powell and immediately transform it into text. We want to create an application that updates hawkish or dovish sentiment sentence by sentence as the speech unfolds. Such an application could eventually serve traders, journalists, and policy analysts who need immediate interpretation of Fed communication.

Second, the largest opportunity for improvement lies in developing a custom model called FedBERT. Rather than relying on FinBERT (which is trained on financial news and earnings reports) we can fine-tune a BERT-style transformer on roughly ten-twenty thousand Federal Reserve speeches, statements, testimonies, and transcripts. Each sentence would be labeled as hawkish, dovish, or neutral creating a domain-specific supervised dataset. To speed this up, I would have code that gives an initial score, and then go through them. If divided up into 3-4 colleagues it could take only a weekend.

The best approach would involve collecting speeches from press conferences, minutes, statements, and regional Fed remarks, labeling sentences using the current lexicon and stance system as a first pass, performing manual correction, and then fine-tuning a pretrained model such as bert base uncased or roberta base on this labeled dataset. A properly trained FedBERT would internalize central bank wording patterns, understand policy context, and outperform FinBERT by a wide margin on this task. This is the natural next step in transforming the model from a rule-guided hybrid into a fully domain-adapted deep learning system.

Overall, the project shows that understanding central bank communication requires both nuance and economic expertise. The progress so far provides a strong foundation for real-time applications and for developing a purpose-built transformer model that captures the subtleties of monetary policy language.